

FINANCIAL DATA ANALYTICS REPORT

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Methodology: CRISP-DM

Executive Summary

This report analyzes financial performance using the Microsoft Financial Sample dataset. The objective of the analysis is to evaluate revenue, expenses, and profitability across different product categories, regions, and customer segments in order to identify opportunities to improve budgeting performance and reduce costs.

The analysis shows that the Paseo product generates the highest revenue and profit, while the United States is the strongest performing market. The Government segment also contributes the highest revenue among customer segments. Additionally, the analysis revealed that sales and profits peak toward the end of the year, particularly in October, December, November, and September.

Several strategic recommendations are proposed including prioritizing high-performing products, optimizing discount strategies, increasing investment in high-revenue markets, reducing operational costs, and allocating higher budgets during peak sales months.

1. Business Understanding

Organizations rely on financial data analytics to monitor performance, manage costs, and support strategic budgeting decisions. In this project, financial sales data was analyzed to identify patterns and trends related to revenue generation, expenses, and profitability.

Problem Statement:

The company needs to determine which products and regions generate the highest revenue and profit while identifying areas where operational costs can be reduced to improve overall financial performance.

Key Business Questions:

- Which product generates the highest revenue?
- What are the total revenues, expenses, and profits across product categories and regions?
- Where can cost-saving opportunities be identified?

2. Data Understanding

The dataset used in this project is the Microsoft Financial Sample dataset, which contains financial sales information across several countries and product categories.

Key variables included in the dataset:

- Segment (customer type)
- Country (sales region)
- Product
- Discount Band
- Units Sold
- Manufacturing Price
- Sale Price
- Gross Sales
- Discounts
- Sales
- Cost of Goods Sold (COGS)
- Profit
- Date and Year

3. Data Preparation

Several data cleaning steps were performed to ensure accuracy and consistency.

- Checked for duplicate records and found none.
- Standardized text values in Segment and Country columns using the PROPER() function.
- Removed row 11 which contained multiple missing dependent values.
- Replaced missing categorical values with the most frequent values in their respective columns.
- Used descriptive statistics to identify outliers and replaced missing numerical values with the median where appropriate.
- Calculated missing Sales values using the formula: $\text{Sales} = \text{Gross Sales} - \text{Discounts}$.
- Created a new column called Profit Margin using the formula $\text{Profit} \div \text{Sales}$.

4. Data Modeling

Pivot tables and pivot charts were created to summarize financial performance across products, regions, segments, discount levels, and time trends.

Product analysis showed that Paseo generated the highest revenue (~\$33M), followed by VTT and Velo. Carretera generated the lowest revenue.

Country analysis showed that the United States generated the highest revenue, followed by Canada, France, and Germany, while Mexico generated the lowest.

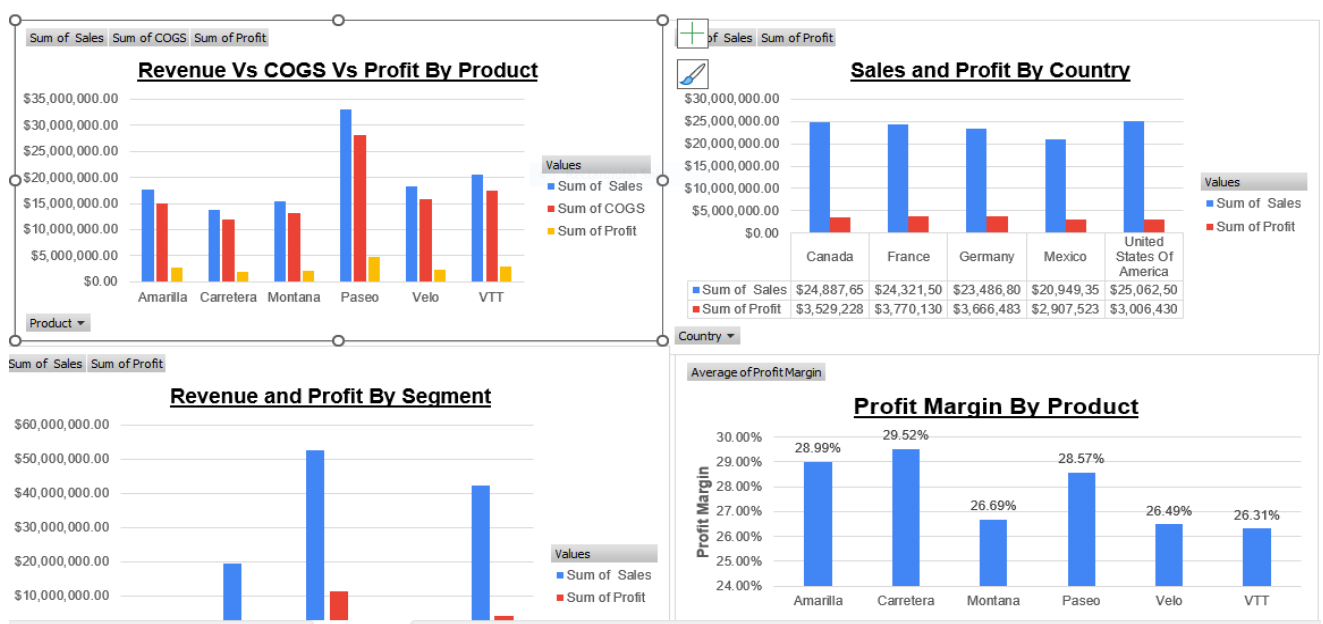
Segment analysis showed that the Government segment generated the highest revenue and profit, while the Enterprise segment recorded lower profitability.

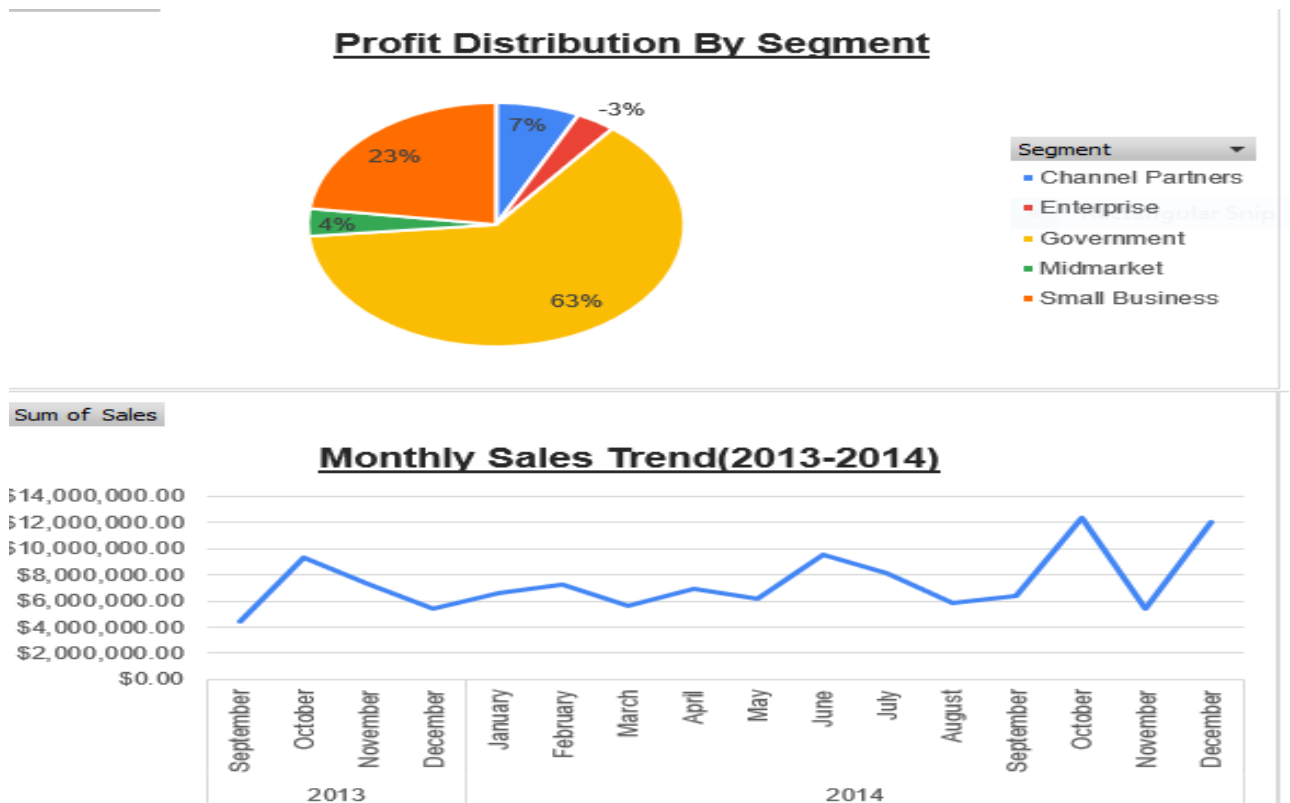
Discount band analysis revealed that low or no discounts maintain stronger profit margins while high discounts significantly reduce profitability.

Monthly sales trends revealed strong seasonal patterns. The highest sales occurred in October, December, November, and September, while March, May, and August recorded the lowest sales.

Profit trends followed a similar pattern with October generating the highest profit.

Financial summary metrics from the dashboard show total revenue of \$118,707,810, total COGS of \$101,828,013, and total profit of \$16,879,797 with an overall profit margin of approximately 27.8%.





5. Evaluation

The analysis revealed several important insights. Paseo is the most profitable product and should be prioritized in future sales strategies. The United States represents the strongest market while the Government segment generates the highest profits.

The analysis also shows that excessive discounting reduces profitability and that sales peak during the final quarter of the year.

6. Recommendations

- Focus marketing and distribution efforts on high-performing products such as Paseo, VTT, and Velo.
- Implement controlled discount strategies to protect profit margins.
- Expand marketing and partnerships in high-revenue markets such as the United States, Canada, and France.

- Reduce operational costs by improving manufacturing efficiency and supply chain management.
- Review pricing strategies for the Enterprise segment to improve profitability.
- Allocate higher marketing and sales budgets during peak months such as October, December, November, and September.
- Introduce promotional strategies during low-sales months such as March, May, and August.

7. Deployment

An Excel dashboard was developed to visually present key financial metrics including revenue by product, revenue by country, revenue by segment, discount analysis, and monthly sales and profit trends.

The dashboard enables stakeholders to quickly interpret financial performance and make data-driven decisions for future budgeting and strategic planning.